

HYPERTENSION: DIAGNOSIS

Check blood pressure (BP)

- **Position patient:** ask patient to remove tight clothing covering upper arm. Seat with back against chair, both feet flat on floor and arm supported at heart level. Patient to avoid talking during reading.
- **Position cuff** correctly and ensure appropriate size: if obese (mid-upper arm circumference is ≥ 33 cm), use a large BP cuff.
- **Measure and record** systolic BP (SBP) and diastolic BP (DBP). Take at least two readings 1-2 minutes apart.
 - If first time BP measurement, or readings differ by > 5 mmHg, or if BP $\geq 180/110$, take a third reading. If able, use average of last 2 readings to interpret BP measurement, otherwise use lowest BP reading.
- If taking BP manually, SBP is the first appearance of sound. DBP is the disappearance of sound.
- If patient is pregnant, interpret reading $\rightarrow$ 159.

Give urgent attention to the patient with BP $\geq 180/110$ (SBP ≥ 180 and/or DBP ≥ 110) and any of:

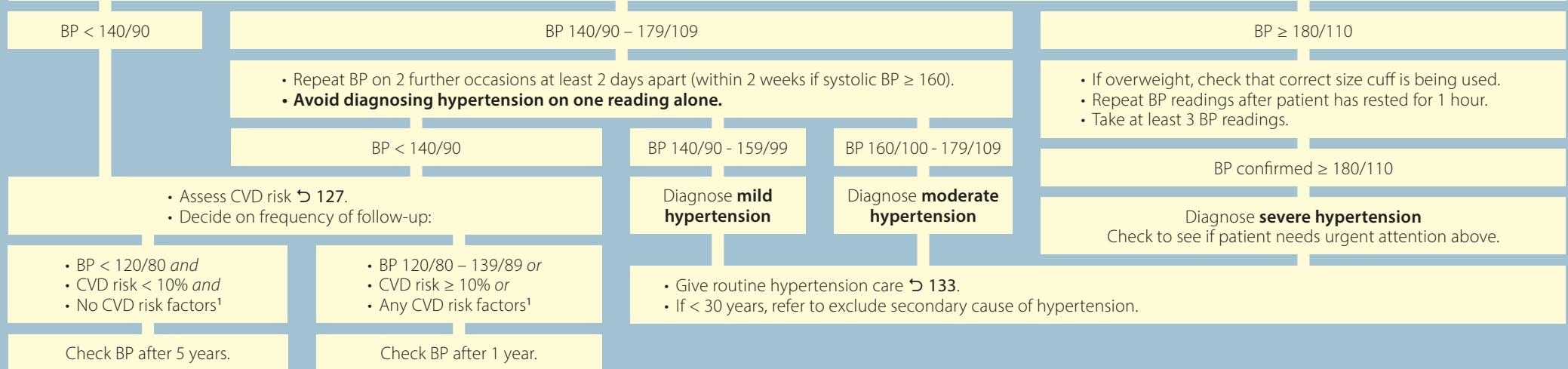
- Visual disturbances
- Dizziness
- Confusion
- Severe headache
- Chest pain $\rightarrow$ 37.
- Difficulty breathing worse on lying flat or with leg swelling $\rightarrow$ 135.
- Sudden weakness on 1 or both sides, vision problems, dizziness, difficulty speaking or swallowing $\rightarrow$ 136.

Manage and refer:

- If BP $\geq 180/130$ (SBP > 180 and/or a DBP > 130) with symptoms listed above, treat as **hypertensive emergency**: give single dose **amlodipine** 10mg orally. Avoid short-acting nifedipine as it may drop the BP too quickly, causing a stroke. Refer urgently.
- If dizzy or faint after treatment, lie patient down. If BP $< 160/100$, raise legs.

Approach to interpreting a BP in a patient not needing urgent attention

If systolic BP and diastolic BP fall into 2 different categories, use the higher blood pressure reading to classify further.



¹CVD risk factors include smoking, diabetes, BMI > 25 , waist circumference > 80 cm (woman) or 94cm (man), cholesterol > 5.2 , parent/sibling with early onset CVD (man < 55 years or woman < 65 years).